Atrial fibrillation and flutter
1. Overview

Based on NICE NG196, 2021

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Detection

If AF suspected, NICE recommends we look for an irregular pulse. This includes those who present with:

- Palpitations/chest discomfort.
- Syncope/dizziness.
- Breathlessness.
- Stroke/TIA.

ADMIT if someone with new/known AF presents acutely unwell/decompensated fast AF/haemodynamically unstable.

This is a case-finding approach: identifying high-risk individuals who you suspect to be at increased risk of AF. **NICE does NOT recommend systematic population screening or opportunistic screening**. The evidence around this is discussed in more detail in our main article on AF.

Investigations

ECG: to confirm diagnosis (24h tape/event recorder if paroxysmal AF suspected). 12-lead ECG should be used until it is clearer whether the newer technologies are sufficiently accurate for diagnosis.

Bloods: FBC, U&Es, TSH and LFTs (NICE recommends no blood tests (!), but these seem sensible, and are needed if starting a DOAC too).

Echo: **NOT routinely indicated**. Do if:

- Structural or functional heart disease (e.g. murmur, heart failure).
- Cardioversion planned (echo can indicate likelihood of success, in part based on size of left atrium).
- Need better stroke risk stratification (e.g. high stroke risk but also high bleeding risk).

Management: think AC for AF!

A – AVOID STROKE		C – CONTROL STRATEGY
Assess stroke and bleeding risk, and anticoagulate if appropriate See section 2	**Patient understanding crucial: important treatment choices to make**	**Decide whether rate or rhythm control strategy needed** See section 3

When to refer

- **Immediately if acutely unwell (fast AF or compromise, e.g. heart failure, hypotensive).**
- If rhythm control strategy to be used (see page 3).
- At any stage if treatment fails to control symptoms and specialised management is needed. Referral should be within 4 weeks of failed treatment/recurrence of AF after cardioversion.

Review annually

For those on anticoagulants: reassess ORBIT and need for (and quality of) anticoagulation at least annually, more frequently if clinical event occurs (e.g. bleed or embolic event).

- **If on DOAC:** check renal function using creatinine clearance (calculated using Cockcroft-Gault formula) and ensure on appropriate dose if renal impairment (MHRA 2019;13:3).
- **If on warfarin:** check time in therapeutic range. Aim for 65% in past 6m (for more info, see main article). Remember, poor compliers will have poor control with DOACs (short half-lives) but you won't know!

If NOT on anticoagulation: NICE says reassess CHA_2DS_2Vasc at age 65y *or* if they develop any of the following: diabetes, heart failure, coronary heart disease, peripheral arterial disease, stroke/TIA or systemic emboli (we note that hypertension and turning 75y are not on this list; however, they give you an extra CHA_2DS_2Vasc point, so recalculation may be indicated then too).

If not on anticoagulation because of bleeding risk/other reason, review stroke and bleeding risks annually.

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2. Avoid stroke: assess stroke/bleeding risk

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Assess stroke & bleeding risk using CHA_2DS_2Vasc & ORBIT: anticoagulate if needed

This applies to all with non-valvular AF, paroxysmal AF or flutter, and those who have been cardioverted but remain at risk of further arrhythmia (presumably, cardiologists to advise on latter group).

Assessing stroke risk with CHA_2DS_2Vasc

- **In most people, the benefits of anticoagulation outweigh the risk.**
  - Falls rarely cause major haemorrhage, and risk of falls alone should not be a reason to withhold anticoagulants.
  - Age alone should not be a reason to withhold anticoagulation.
- If high bleeding risk, benefits of anticoagulation may not outweigh the bleeding risks. Monitor bleeding risks carefully.
- **CHA_2DS_2Vasc and ORBIT are helpful, but patient-specific factors (e.g. comorbidities, patient preference) should also be taken into account. Don't make your decision just based on score alone!**

CHA_2DS_2Vasc	Score
CHF	1
Hypertension	1
Age	65–74y: 1 ≥75y: 2
Diabetes	1
Stroke/TIA	2
Sex (female)	1
Vascular disease	1

Interpreting CHA_2DS_2Vasc score

$CHA_2DS_2Vasc = 0$	$CHA_2DS_2Vasc = 1$	$CHA_2DS_2Vasc \geq 2$
Do NOT offer anticoagulation (risk of embolic event <1%)	**MALE: Consider anticoagulation FEMALE: Do NOT offer anticoagulation (risk low)**	**Offer anticoagulation (stroke risk >2%/y)**

Which anticoagulant?

- **DOACs are first line (any DOAC)** (because more effective than warfarin). Use warfarin if DOACs contraindicated (e.g. low creatinine clearance, antiphospholipid syndrome) or not tolerated.
- If anticoagulation contraindicated/not tolerated, consider left atrial appendage occlusion (see main *Atrial fibrillation* article). Do NOT offer aspirin for stroke prevention in AF.
- **How promptly to anticoagulate?** NICE gives no timescale, but once AF is recognised, we suggest prompt anticoagulation as each day without carries a risk of stroke.
- **For those on warfarin because they were diagnosed with AF some years ago:** discuss the option to switch to a DOAC at next routine review.

If anticoagulants being considered, assess bleeding risk with ORBIT

Although bleeding risk scores may occasionally be used as a reason not to offer anticoagulation, they should usually be used to identify and manage modifiable risk factors, rather than be a reason not to anticoagulate.

Assess, address and monitor modifiable factors that increase bleeding

- Harmful alcohol consumption.
- Drugs that increase bleeding risk, including antiplatelets, NSAIDs and SSRIs.
- Uncontrolled hypertension.
- Reversible causes of anaemia.
- Poor control of INR in those on warfarin.

Interpreting ORBIT score

0–2 = low risk (2.4 bleed/100 patient years).
3 = medium risk (4.7 bleeds/100 patient years).
≥4 = high risk (8.1 bleeds/100 patient years).

ORBIT bleeding risk score	Score
Man with Hb <13 or haematocrit <40% or Woman with Hb <12 or haematocrit <36%	2
Age ≥75y	1
Bleeding history (GI/intracranial bleed or haemorrhagic stroke)	2
eGFR<60	1
On antiplatelets	1

Undetectable AF

NICE reminds us that anticoagulation should not be stopped just because AF is no longer detected (especially in those paroxysmal AF).

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3. Control strategy: rate or rhythm control?

Based on NICE NG196, 2021

Decide on whether rate or rhythm control is most appropriate

- **This section does NOT apply to paroxysmal AF (see next page).**
- **If acutely unwell: ADMIT.**
- **If onset in past 48h, cardioversion (electrical or chemical) can be offered without anticoagulation:** discuss with specialist today to decide whether suitable for this (if >48h since onset, must anticoagulate for at least 3w before considering for cardioversion). (NOTE: this assumes there is a clear 'onset', such as a sudden heave in the chest and a sense of palpitations since then.)

Rate or rhythm control?

RATE CONTROL first line for everyone UNLESS meets criteria for rhythm control

The RHYTHM control criteria are:

- **New-onset AF** (new onset, not a newly-discovered incidental finding).
- **AF with reversible cause** (e.g. triggered by pneumonia).
- **Heart failure caused by AF.**
- **Atrial flutter where the patient may be suitable for ablation to restore sinus rhythm** (cardiologists tell us that flutter can be harder to control with drugs and may be more amenable to ablation).
- **Rhythm control more suitable based on clinical judgment**. NICE doesn't define this group but some friendly cardiologists suggested this might be:
  - Younger patients.
  - Those with a structurally-normal heart (difficult in primary care as we are told not to do an echo unless we suspect structural/functional disease!).

Cardiologists tell us that the more normal the atria, the greater the chance of keeping someone in sinus rhythm. In those who have been in AF for some time, or have structurally-abnormal hearts, this chance diminishes because of atrial remodeling.

NONE of the above: RATE control

(the majority)

AIM: pulse <110bpm and symptom free

Manage in primary care:
MONOTHERAPY WITH
any beta-blocker (except sotalol)
or
rate-limiting calcium channel blocker (diltiazem or verapamil) (avoid CCB in heart failure) (diltiazem use is off-label).

Use digoxin monotherapy only if no/very little exercise or unsuitable for other options (only controls rate at rest).

If monotherapy ineffective:
TRY DUAL THERAPY with any 2 of:
beta-blocker/diltiazem/digoxin.
Do NOT use amiodarone for long-term rate control (significant, often irreversible, side-effects, including thyroid, lung and nerve damage).
If dual therapy ineffective, refer for rhythm control/ablation.

ANY of the above: RHYTHM control

(the minority)

AIM: to restore sinus rhythm

REFER FOR CARDIOVERSION
Electrical cardioversion for most
(chemical cardioversion may be used if AF present for <48h).

After cardioversion, some will be given drugs for long-term rhythm control. Beta-blockers (not sotalol) used first line. Dronedarone second line if specific NICE criteria met (TA197). Amiodarone may be used if LV impairment or heart failure or for a limited time (1–12m) after electrical cardioversion (may aid remodelling of atrial pathways).

If cardioversion fails/unsuitable, consider left atrial ablation (percutaneous or surgical).
Patient needs to understand that ablation is not always successful and, if successful, may not be long-lasting.
After ablation, antiarrhythmics may be given for 3m to try and prevent recurrence of AF.
Pace and ablate (AV node ablation + pacemaker fitted): used if LV dysfunction thought to be caused by high ventricular rate, or in paroxysmal AF if other strategies have failed.

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4. Management of PAROXYSMAL AF
Based on NICE NG196, 2021 and BMJ 2021;375:e058568

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How common is paroxysmal AF?
- Prevalence increases with age:
  - In those aged 75y, prevalence is 12–14%.
  - In those ≥85y, it is 18%.

Paroxysmal AF is easily missed because it may be asymptomatic. The commonest symptoms (in decreasing order of frequency) are:
- Palpitations.
- Dyspnoea on exertion.
- Fatigue.
- Lightheadedness.
- Dyspnoea at rest.
- Reduced exercise tolerance.
- Chest tightness.
- Syncope.

Diagnosis/investigations of paroxysmal AF
- May need to do 24h tape or ambulatory ECG or event recorder to detect paroxysmal AF, so a normal ECG does not rule this out.
- Otherwise, investigations are the same as for ongoing AF.

Management of paroxysmal AF

A - AVOID STROKE: assess stroke and bleeding risk using CHA_2DS_2Vasc and ORBIT (see page 2)
- Start anticoagulation, if indicated.
- FREQUENCY of episodes may affect risk (the greater the frequency, the higher the risk), but remember that some/many episodes may be silent.

C- CONTROL strategy: REFER for specialist to decide control strategy
Specialist to decide whether to aim for long-term rhythm control or to take a 'pill in the pocket' approach.
Factors that influence that decision:

Infrequent but symptomatic paroxysms, especially if known precipitants (caffeine, alcohol)
AND
no history of LVSD/valve disease/IHD
AND
systolic BP >100
AND
resting pulse rate >70bpm

'YES' TO ALL OF THE ABOVE:
'PILL IN THE POCKET'
Aim to abort AF when it occurs
(this will be the minority)
- Suitable for those on no or low-dose maintenance doses of antiarrhythmics (beta-blocker or flecainide often used).
- Aim is to abort the attack by taking a single dose of drug (e.g. flecainide) at onset of symptoms, and avoid the need for A&E visits/admission.

'NO' TO ANY OF THE ABOVE:
LONG-TERM RHYTHM CONTROL
Aim to hold in sinus rhythm (this will be the majority)
- **Beta-blockers (not sotalol) are used first line.** Alternatives include:
  - Amiodarone (if heart failure/LV impairment).
  - Sotalol (care in renal impairment/low BMI).
  - Dronedarone.
- **Consider left atrial ablation if symptomatic paroxysmal AF if drug treatment unsuccessful/not tolerated/contraindicated.** Patient needs to understand that left atrial ablation is not always successful and, if successful, may not be long-lasting.